



STA 4322
Introduction to Mathematical Statistics I
84854 001
Fall 2026

Course Information

Meeting Days, Time and Location: MW 1100 - 1215 ISA3048

Delivery Method: Classroom

Instructor Name

Xiankui Yang

Instructor Information

Name: Xiankui Yang

Email: xiankuiyang@usf.edu

Office Hours

MW 9:30 am - 10:30 am CMC 028

University Course Description

Common distribution functions, sampling distributions, statistical point and interval estimation, properties of estimators, hypothesis testing.

Course Requisites

Prerequisite(s):

STA 4442 Minimum Grade: C

Corequisite(s):

Course Purpose

This course focuses on the theory of mathematical statistics and its applications.

Student Learning Outcomes

1. Finding distributions of random variables
2. Computing numerical characteristics of the probability distributions
3. Practical applications of the most frequently used probability distributions
4. Finding the distribution of a transformed random variable
5. Applying the CLT and other asymptotic results for performing the approximate probability computations and proofs.
6. The sampling distributions of the sample mean and variance
7. Finding the point estimators using the method of moments and the method of maximum likelihood
8. Using the mean squared error for assessing the statistical properties of the point estimators
9. Using the pivotal methods for constructing the confidence intervals for the population parameters.
10. Computing different types of interval estimators of the population parameters
11. Testing statistical hypotheses for the population parameters

Course Objectives

- Understand common probability distribution functions and their applications.

- Apply sampling distributions in statistical inference.
- Construct and interpret point and interval estimates for population parameters.
- Evaluate the properties of statistical estimators.
- Perform and interpret hypothesis tests for statistical decision-making.

Required Texts and/or Readings and Course Materials

Mathematical Statistics with Applications in R by Ramachandran & Tsokos, Academic Press, third edition.

Grading Scale

The grading scale is as follows:

A+ 97 - 100%, A 93 - 96%, A- 90 - 92%, B+ 87 - 89%, B 83 - 86%, B- 80 - 82%,

C+ 77 - 79% C 73 - 76% C- 70 - 73%, D+ 67 - 69%, D 63 - 66% , D- 60 - 62%,

F less than 60%.

Grade Categories and Weights

Assessment	Percent of Final Grade
Attendance	10%
Homework	10%
Midterm Exam I	25%
Midterm Exam II	25%
Final Exam	30%
	100%

Instructor Feedback Policy & Grade Dissemination

Homework Policy:

- Homework problems offer practices for applying lecture materials; it is an important and most effective way for mastering the course material.
- There will be **six homework assignments**. All homework will be posted on Canvas with their due dates written on the assignment sheets.
- All homework should be scanned and submitted in Canvas. No late homework or homework sent via email will be accepted. The lowest score among the six graded homework assignments will be dropped from the final grade calculation.
- You are encouraged to work in groups on homework problems; however, you must write your own answers. Copying others' answers is not a good way to learn.

Exam Policy:

- There will be **two midterm exams** and **one final exam**. All the exams will be closed-book.
- The midterm exams will be written in class. The coverage of the exam will be announced a week before the scheduled exam date. The final exam will be cumulative.
- Makeup exams will be given only if students contact the instructor and get approval prior to the scheduled exam. In the event of an unforeseen emergency that prevents the student from taking the exam as scheduled, the student must provide documentation to the instructor before a makeup exam can be arranged. However, different questions will be used for makeup

exams.

Attendance Policy:

- Students are expected to attend class regularly. Absences due to illness, emergencies, or other reasonable circumstances may be excused with appropriate notice or documentation.
- Each student is allowed **three unexcused absences** during the semester without penalty.
- Attendance will be recorded using a **sign-in sheet** in class. Students are responsible for signing their own names on the attendance sheet.
- Signing in for another student is not allowed and may be considered a violation of the course attendance policy.

Major Topics and Course Schedule

Chapter 2: Only moment generating function.

Chapter 3: Additional topics in probability

Chapter 4: Sampling distribution

Chapter 5: Statistical estimation

Chapter 6: Hypothesis testing

Homework assignments and in-class exercises

Additional lecture notes and R demonstrations (as provided by the instructor)

USF Core Syllabus Policies

USF has a set of central policies related to student recording class sessions, academic integrity and grievances, student accessibility services, academic disruption, religious observances, academic continuity, food insecurity, pregnancy and related conditions, and sexual harassment that **apply to all courses at USF**. Be sure to review these online:

<https://www.usf.edu/provost/faculty-success/resources-policies-forms/core-syllabus-policy-statements.aspx>

Student Recordings

Florida Statute 1004.097 Free Expression on Campus

Students may, without prior notice, record video or audio of a class lecture for a class in which the student is enrolled for their own personal, educational use (USF Policy 10-048). A class lecture is defined as a formal or methodical oral presentation as part of a university course intended to present information or teach enrolled students about a particular subject. Recording class activities other than class lectures, including but not limited to lab sessions, student presentations (whether individually or part of a group), class discussion, clinical presentations such as patient history, academic exercises involving student participation, test or examination administrations, field trips, private conversations between students in the class or between a student and the faculty member is prohibited. Recordings may not be used as a substitute for class participation and class attendance and may not be published or shared without the written consent of the faculty member. Failure to adhere to these requirements may constitute a violation of the USF Student Conduct Code.

Course Policies: Grades

First Day Attendance Policy:

Attendance at class meetings is essential to academic success. Full time

attendance to every class is strongly recommended. **First class attendance is required** by the university for students who registered prior to the first day of the class.

Late Work Policy:

No late homework or homework sent via email will be accepted.

Make-up Exams Policy:

Makeup exams will be given only if students contact the instructor and get approval **prior** to the scheduled exam. In the event of an unforeseen emergency that prevents the student from taking the exam as scheduled, the student must provide documentation to the instructor before a makeup exam can be arranged. However, **different** questions will be used for makeup exams.

Final Examinations Policy:

All final exams are to be scheduled in accordance with the University's final examination policy.

Bonus Points Policy:

Bonus point opportunities may be offered during class at the instructor's discretion. These points may be earned through in-class activities, practice problems, or participation. Bonus points are not guaranteed and cannot be made up if a student is absent.

Grades of "Incomplete"

The current university policy concerning incomplete grades will be followed in this course. For undergraduate courses: An "I" grade may be awarded to a student only when a small portion of the student's work is incomplete and only when the student is otherwise earning a passing grade. The time limit for removing the "I" is to be set by the instructor of the course. For undergraduate students, this time limit may not exceed two academic semesters, whether or not the student is in residence, and/or graduation, whichever comes first. For graduate students, this time limit may not exceed one academic semester. "I" grades not removed by the end of the time limit will be changed to "IF" or "IU," whichever is appropriate.

Campus Free Expression

It is fundamental to the University of South Florida's mission to support an environment where divergent ideas, theories, and philosophies can be openly exchanged and critically evaluated. Consistent with these principles, this course may involve discussion of ideas that you find uncomfortable, disagreeable, or even offensive.

In the instructional setting, ideas are intended to be presented in an objective manner and not as an endorsement of what you should personally believe. "Objective" means that the idea(s) presented can be tested by critical peer review and rigorous debate, and that the idea(s) is supported by credible research.

In this course you may be asked to engage with complex ideas and to demonstrate an understanding of the ideas. Understanding and engaging with an idea does not require you to believe it or to agree with it.

Course Policies: Student Communication

WhatsApp, GroupMe, and Student-to-Student Communication: While students may use digital communication tools (WhatsApp, GroupMe, etc.) to communicate with fellow students, it is important to remember that academic integrity policies still apply in these environments. Informing others about the contents of tests is prohibited by the official regulation, as is receiving unauthorized information about an examination. Students are expected and required to immediately report instances of such violations to the instructor.

Course Policies: Technology and Media

Email: Emails will be used to make announcements and answer class related questions. I'll try to respond as quickly as I can during week days. However, if a student plans to ask questions or send any inquiry via emails, please make sure enough time is allowed to respond.

Canvas: Course materials including class notes, homework assignments, and other supporting materials will be posted on Canvas. It is the students' responsibility to login to Canvas several times per week to check for updates.

Classroom Devices: Students can use calculators and laptops in classes. However, no tape recorders, cameras, or other video/audio & technology devices are allowed in the classroom.

Course Policies: Student Expectations

Health and Wellness

Your health is a priority at the University of South Florida. We encourage members of our community to look out for each other and to reach out for help if someone is in need. If you or someone you know is in distress, please make a referral at www.usf.edu/sos so that the Student Outreach & Support can contact and provide helpful resources to the student in distress. A 24-hour licensed mental healthcare professional, offered through the counseling center, is available by phone at 813-974-2831, option 3. Please remember that asking for help is a sign of strength. In case of emergency, please dial 9-1-1.

Title IX Policy

USF is committed to fostering a safe and respectful learning environment, free from sex discrimination and sexual harassment, in accordance with Title IX and USF Policy 0-004. If a student shares an experience of sexual harassment, sexual assault, stalking, or relationship violence with a USF employee (including faculty), that employee will notify the Title IX Office. This allows the Title IX Office to send the student an email with information about support resources and options—if it is safe to do so. Students are not required to respond to this email. The Title IX Office is a private resource. Information is shared only on a need-to-know basis to provide students with support measures. To speak to a confidential resource, students may contact USF Center for Victim Advocacy at 813-974-5757. Confidential resources do not notify the Title IX Office unless the student requests it. For more information about Title IX, available resources, or to make a report, please visit www.usf.edu/title-ix.

Discrimination / Harassment

Discrimination or harassment of students based on protected characteristics is prohibited by Title VI, the Florida Educational Equity Act, and [USF Policy 0-007](#). Protected characteristics include race, color, national origin, shared ancestry, religion, age, sex, disability, genetic status, marital status, and veteran status. If a student shares an experience of harassment or discrimination with a USF employee (including faculty), that employee will notify the Student Equal Opportunity Office. This allows the Student EO Office to send the student an email with information about support resources and options—if it is safe to do so. Students are not required to respond to this email. The Student EO Office is a private resource. Information is shared only on a need-to-know basis to provide students with support measures. For more information about Student EO, available resources, or to make a report,

please visit www.usf.edu/student-eo

Course Hero / Chegg Policy

The USF Policy on Academic Integrity specifies that students may not use websites that enable cheating, such as by uploading or downloading material for this purpose. This does apply specifically to Chegg.com and CourseHero.com – almost any use of these websites (including uploading proprietary materials) constitutes a violation of the academic integrity policy.

Professionalism Policy

Per university policy and classroom etiquette; mobile phones, iPods, etc. must be silenced during all classroom and lab lectures. Those not heeding this rule will be asked to leave the classroom/lab immediately so as to not disrupt the learning environment. Please arrive on time for all class meetings. Students who habitually disturb the class by talking, arriving late, etc., and have been warned may suffer a reduction in their final class grade.

Turnitin.com

In this course, turnitin.com may be utilized. Turnitin is an automated system which instructors may use to quickly and easily compare each student's assignment with billions of web sites, as well as an enormous database of student papers that grows with each submission. Accordingly, you may be expected to submit all assignments in both hard copy and electronic format. After the assignment is processed, as instructor I receive a report from turnitin.com that states if and how another author's work was used in the assignment. For a more detailed look at this process visit <http://www.turnitin.com>.

Netiquette Guidelines

- Act professionally in the way you communicate. Treat your instructors

and peers with respect, the same way you would do in a face-to-face environment. Respect other people's ideas and be constructive when explaining your views about points you may not agree with.

- Be sensitive. Be respectful and sensitive when sharing your ideas and opinions. There will be people in your class with different linguistic backgrounds, political and religious beliefs or other general differences.
- Proofread and check spelling. Doing this before sending an email or posting a thread on a discussion board will allow you to make sure your message is clear and thoughtful. Avoid the use of all capital letters, it can be perceived as if you are shouting, and it is more difficult to read.
- Keep your communications focused and stay on topic. Complete your ideas before changing the subject. By keeping the message on focus you allow the readers to easily get your idea or answers they are looking for.
- Be clear with your message. Avoid using humor or sarcasm. Since people can't see your expressions or hear your tone of voice, meaning can be misinterpreted.

Email and Discussion Board Guidelines

- Use the subject line effectively by using a meaningful line of what your email or discussion is about.
- Keep your emails and postings related to the course content. You should not post anything personal on a discussion board unless it is requested by the instructor.
- Any personal, course or confidential issues should be directly communicated to the instructor via email. The discussion boards are public spaces; therefore, any issues should not be posted there.

End of Semester Student Evaluations

All classes at USF make use of an online system for students to provide feedback to the University regarding the course. These surveys will be made available at the end of the semester, and the University will notify you by email when the response window opens. Your participation is highly encouraged and valued.

Food and Drink Policy (for in-campus sessions only).

Please adhere to the firm policy of no beverages (other than bottled/capped water), food, tobacco products, or like items in the classroom. Your understanding of the necessity for this policy and cooperation will be greatly appreciated. This policy will be strictly enforced.

Generative AI

Students may use generative AI tools to support learning and study. However, unless otherwise specified by the instructor, **all graded work must be completed independently and must reflect the student's own understanding**. Students are responsible for the accuracy and appropriate use of any AI-generated content and must comply with the University's Academic Integrity Policy.

Learning Support and Campus Offices

Academic Accommodations

Students with disabilities are responsible for registering with Student Accessibility Services (SAS) in order to receive academic accommodations. For additional information about academic accommodations and resources, you can visit the SAS website.

[SAS website for the Tampa campus and Sarasota-Manatee Teachout site.](#)

[SAS website for the St. Pete campus.](#)

Academic Support Services

The USF Office of Student Success coordinates and promotes university-wide efforts to enhance undergraduate and graduate student success. For a comprehensive list of academic support services available to all USF students, please visit the [Office of Student Success website.](#)

Canvas Technical Support

If you have technical difficulties in Canvas, you can find access to the Canvas guides and video resources in the “Canvas Help” page on the homepage of your Canvas course. You can also contact the help desk by calling 813-974-1222 in Tampa or emailing help@usf.edu.

[USF IT Services](#)

Center for Victim Advocacy

The [Center for Victim Advocacy](#) empowers survivors of crime, violence, or abuse by promoting the restoration of decision making, by advocating for their rights, and by offering support and resources. Contact information is available online.

Counseling Center

The Counseling Center promotes the wellbeing of the campus community by providing culturally sensitive counseling, consultation, prevention, and training that enhances student academic and personal success. Contact information is available online.

[Counseling Center website for the Tampa campus and Sarasota-Manatee Teachout site.](#)

[Counseling Center website for the St. Pete campus.](#)

Tutoring

The Tutoring Hub offers free tutoring in several subjects to USF undergraduates. Appointments are recommended, but not required. For more information, email

asctampa@usf.edu.

[Tutoring website for the Tampa campus and Sarasota-Manatee Teachout site.](#)

[Tutoring website for the St. Pete campus.](#)

Writing Studio

The Writing Studio is a free resource for USF undergraduate and graduate students. At the Writing Studio, a trained writing consultant will work individually with you, at any point in the writing process from brainstorming to editing. Appointments are recommended, but not required. For more information or to make an appointment, email: writingstudio@usf.edu.

[Writing studio website for the Tampa campus and Sarasota-Manatee Teachout Site.](#)

[Writing studio website for the St. Pete campus.](#)

Student Ombuds Office

Florida Statute 1006.51 and Florida Board of Governors Regulation 6.011 require that all public institutions provide ombuds services to students. The Student Ombuds Office at USF is a confidential, impartial, informal, and independent resource for students experiencing challenges related to the university. Ombuds can help students safely explore options, clarify processes, and consider next steps. [Learn more or make an appointment at usf.edu/student-ombuds.](https://usf.edu/student-ombuds)

Important Dates to Remember

First Day Attendance: Monday, August 24, 2026

Drop/Add Deadline: Friday, August 28, 2026

Labor Day Holiday: Monday, September 7, 2026 (No Class)

Withdrawal Deadline: Saturday, October 24, 2026

Veteran's Day Holiday: Wednesday, November 11, 2026 (No Class)

Thanksgiving Holiday: Thursday, November 26 & Friday, November 27, 2026

Exam 1: Wednesday, September 23, 2026

Exam 2: Wednesday, November 04, 2026

Final Exam: Monday, December 07, 2026